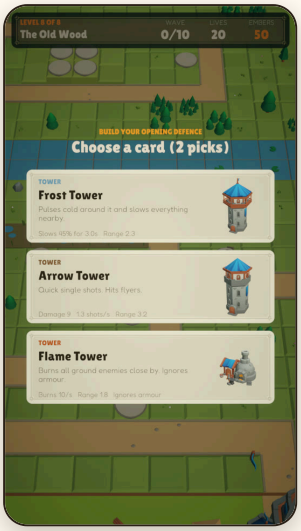
GAME DESIGN DOCUMENT · 0.3.0

Bonemarch

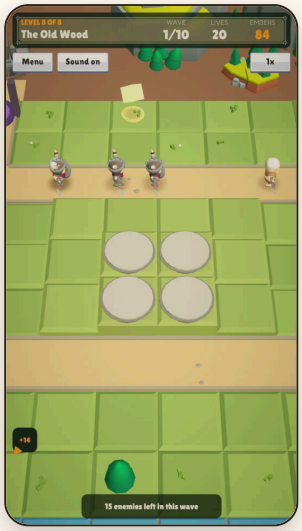
Draft your towers. Hold the gate. A calm 3D tower defense for phones: the strategy is which card to take and where to put it, and each level is a new place on an island of eight.



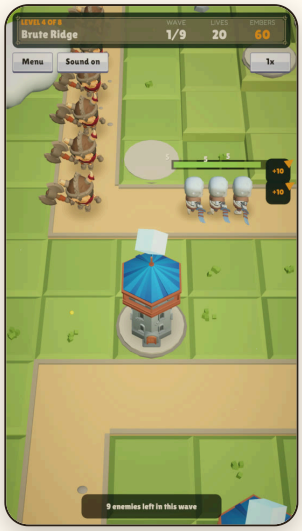
The island map



The draft: two picks from three



A wave on level 8



Brutes march on the towers

At a glance

Genre	Hybrid-casual 3D tower defense with a card draft
Platform	Portrait mobile (Android) and mobile web; Unity 6, URP
Session	One level: 2.5 to 13.5 minutes of fighting (first-time bot), at the player's pace between waves
Content	8 hand-made levels on maps from 9 × 13 to 17 × 22 tiles; 5 towers, 7 enemies, 10 boons, 5 landmarks
Audience	Casual players who like tower defense but bounce off fast, fiddly ones; the audience of 3D card-led defense games

Pillars

Calm, not slow

No timer between waves: the player presses Start. 2× speed is optional. Difficulty comes from new enemies and the map, not from rushing.

The draft is the strategy

No fixed tower menu: two picks from three cards before wave 1, one after every wave. Every run of a level is a different hand.

Every level is a place

Each map has one landmark that changes where towers belong: a bridge, a hill, a fen, a fork, ruins.

Readable at a glance

Enemies read by silhouette (shield, hood, axe, size); a tower's level by its roof colour; pads glow only while a tower is in hand.

RULES

One level, step by step

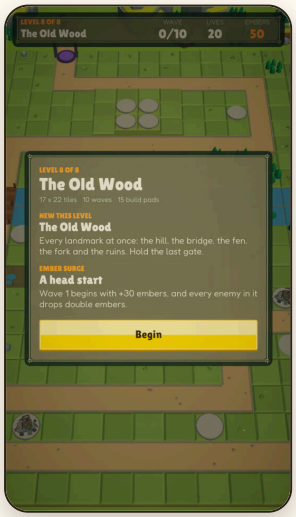
1. **Read the map.** The intro panel names the level's landmark and its new enemy; the camera opens on the gate. Swipe to look around, pinch to zoom.
2. **Draft.** Pick two cards from three. A tower card goes into your hand; a **Twin** card gives two of one tower. The offer always holds at least one tower while there is a free pad.
3. **Build.** Tap a glowing pad. Pads sit beside the road; towers on **high ground** reach 25% farther; a **ruined** pad costs 40 embers to clear first.
4. **Start the wave** when ready. Skeletons walk the road to the gate; each one that arrives costs a life (a Bone Giant five). The player has 20.
5. **Defend.** Towers fire on their own. Kills drop **embers**; tap a tower to upgrade it (30, then 60 embers: 1.6× then 2.4× damage, more range). **Crystals** appear around the map: tap to collect.
6. **Draft again** after every wave: one pick from three, now with **boons** (from level 3). A wave clear pays 8 embers plus 2 for each wave before it.
7. **Clear the last wave** to win. Stars come from lives left: three at 18 or more, two at 10 or more, else one. The next place on the island opens.

> **The Ember Surge.** Wave 1 of every level after the first brings the level's new enemy, with +30 embers up front and double embers for kills, so the player can afford the answer the moment it walks on.

Five towers

TOWER	RANGE	DAMAGE	ATTACKS/S	WHAT IT DOES	FROM LEVEL
Arrow Tower	3.2	9	1.3	Single shots; hits flyers	1
Cannon Tower	2.8	16	0.55	Splash 1.3 tiles; ground only	1
Frost Tower	2.3	5	0.8	Pulse that slows by 45% for 3 s; hits flyers	2
Flame Tower	1.8	10	4	Burns every ground enemy close by; ignores armour	4
Storm Tower	3.0	9	0.7	Lightning that jumps between 3 enemies; hits flyers	5

Range in tiles. Level 2 and 3 towers deal 1.6× and 2.4× damage and reach 0.3 tiles farther a level; a level-3 Frost slows by 55%, and Storm lightning jumps once more a level.



Intro: the landmark and the new enemy



Build: pads glow while a tower is in hand



Tap a tower to upgrade it

RULES

Seven enemies, ten boons

Each level brings one enemy that asks a new question, and the draft holds its answer.

ENEMY	HEALTH	SPEED	ARMOUR	EMBER	WHAT IT DOES	FROM LEVEL
Minion	20	1.0	–	2	The basic walker	1
Shieldbearer	34	0.8	3	3	Each hit does 3 less damage	2
Runner	13	1.9	–	2	Twice as fast, fragile; spawns 0.5 s apart	3
Brute	30	0.9	–	3	Falls apart into two Bone Bits (10 health)	4
Wraith	18	1.2	–	3	Flies: Cannon and Flame can't reach it	5
Necromancer	32	0.85	–	4	Heals the skeletons walking beside it	6
Bone Giant	240	0.45	5	15	Costs 5 lives at the gate; spawns 3 s apart	7

Health before the level's multiplier (level 1 sets 3×, later levels 2× to 9×, growing 25% a wave); speed in tiles a second; armour is taken off every hit except Flame's.

Boons

BOON	EFFECT
Fletcher's Oath	Arrow Towers deal 30% more damage
Black Powder	Cannon blasts are 35% wider
Deep Frost	Frost slows last 50% longer
Dry Kindling	Flame Towers burn 30% hotter
Copper Rod	Storm lightning makes 1 more jump
Hearthstone	Gain 4 lives
Ember Cache	Gain 30 embers now
Steady Hands	All towers attack 15% faster
Watchtower Lens	All towers reach 12% farther
Guild Discount	Upgrades cost 25% less (once)

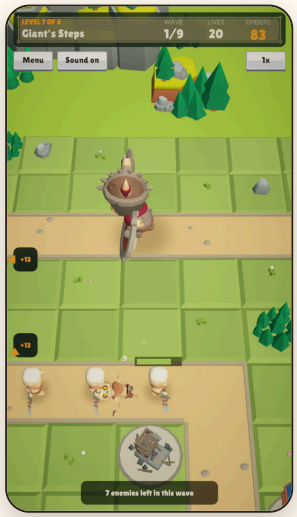
What the bots say about cards

Taking a tower away from the practised bot costs it, in points of win rate: **Storm 10, Frost 10, Cannon 8, Flame 5, Arrow 1**. Every tower has a job; the Arrow's is the air before Storm arrives.

> **Boons are not yet worth a pick.** Each moves the first-time bot by −3 to +2 points, and doubling them all barely changed that: a boon replaces a tower card, and a new tower covering fresh road is worth as much as a percentage on the towers already built. Three options are open: a boon as a free extra pick, towers that cost embers, or boons that change where towers work.



Wraiths float over the fen



A Bone Giant: five lives at the gate



Every tower's hit effect

LEVEL DESIGN

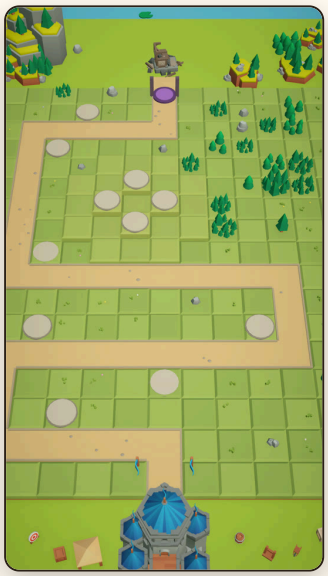
Eight places, one new idea each

Every level adds one enemy and one landmark, and the landmark changes where towers belong. Waves are written by hand: the new enemy in wave 1, a breather after each peak, the biggest wave last.

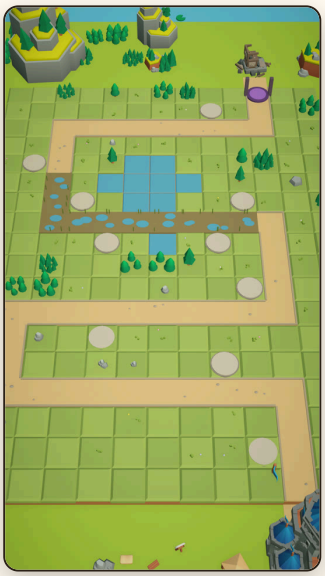
#	LEVEL	MAP	LANDMARK	NEW ENEMY	TOWERS	WAVES	PAD
1	**Mossgate Meadow**	9×13	none: a short road that doubles back twice	Minion	Arrow, Cannon	6	6
2	**Shieldwall Brook**	11×15	the bridge, the only crossing of the brook	Shieldbearer	+ Frost	8	8
3	**Ember Hollow**	11×16	switchbacks: pads between lanes reach two	Runner		8	11
4	**Brute Ridge**	13×17	high ground: four hill pads, +25% range	Brute	+ Flame	9	11
5	**Wraith Fen**	13×18	the fen: marsh slows walkers to 60%	Wraith	+ Storm	9	10
6	**Necromancer's Grove**	15×19	the fork: a side road joins halfway	Necromancer		9	10
7	**Giant's Steps**	15×20	ruined walls: five pads under rubble	Bone Giant		9	11
8	**The Old Wood**	17×22	all of them at once	everything		10	15



2 · Shieldwall Brook



4 · Brute Ridge



5 · Wraith Fen



7 · Giant's Steps

A level is a text file

All eight live in `Levels.txt`, which the game, the tests and the bots all read. The map is drawn in tiles (`#` road, `o` pad, `H` hill, `R` ruin, `~` marsh, `=` bridge, `:` side road); a checker names the line of any mistake, such as a road that touches itself or stops short of the gate.

> **Drafted with AI, redesigned by hand.** The eight maps and waves in this build are first drafts made with an AI assistant from my brief, so the format, the landmark rules and the bots could be built and tested on real levels. The shipped levels are mine to redraw.

```
level Brute Ridge
towers Arrow Cannon Frost Flame
enemy Brute
rules upgrades boons crystals
intro High ground | Towers on the hill ...
map
. . . . . S . . . . . .
. o . . . . # . . o . . .
. . # # # # # . . . . . .
. . # . . H H . . # # # .
...
wave 3 Minion, 6 Brute
wave mixed 8 Minion, 4 Runner, 5 Brute
```

BALANCE

Tuned by bots, read by a designer

A balance runner compiles the game's own rules outside Unity and plays every level 300 times a cell with bots that stand in for kinds of player.

Careless

Any card, built nearest the gate. The floor: reading the game should beat it.

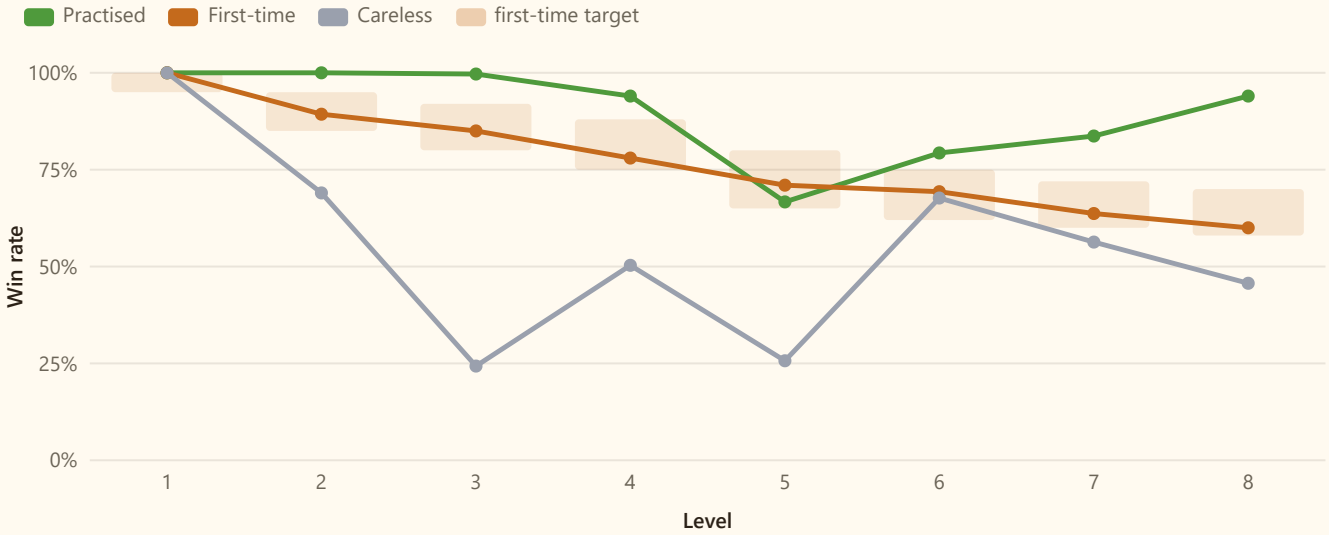
First-time

The card that adds the most damage; heeds the intro's warnings. The target curve is set for this one.

Practised

Plans for the level's threats and builds where towers cover the most road.

Win rate by level



What it found

- Each level's enemy health was fitted to put the first-time bot on its target, from a sure win on level 1 to three in five on level 8.
- After the second pass the counter matters: without it, the practised bot loses 11 to 21 points on levels 3 to 5.
- Level 7's Bone Giants decide it in the last wave (65 of the first-time bot's 109 losses): a set piece to check in a playtest.
- Level 6 rewards reading too little: the careless and first-time bots win about as often. Noted for the redesign.

Hand-made levels fixed the long ones

FIRST-TIME BOT	PHASE 3: GENERATED MAPS	PHASE 6: HAND-MADE DRAFTS
Minutes of fighting, level 7 / 8	24.0 / 36.7	13.5 / 13.5
Careless bot's win rate, level 5	70%	26%
Embers left at the end, level 8 (practised)	761	391

Bots never misread the map or panic: the numbers steer the tuning; a playtest decides.

FEEL, LOOK AND NEXT STEPS

Art, audio, tech and the roadmap

Art direction

- Low-poly 3D in daylight on free CC0 kits (KayKit's medieval set and animated skeletons); the grid reads as chunky land with bevelled tiles and a kerb along the road.
- Towers and enemies are the most saturated things on screen; a tower's roof turns blue, red, then gold as it levels up.
- Every attack shows at its target: arrows with trails, cannon dust and decals, frost rings, flame licks, storm bolts with a flash at each hop. Particles are built in code.
- Blob shadows instead of real-time ones (the web build drops 3D draws with URP shadows on); mild bloom on embers and crystals.

Audio

- Two loops in D major at 74 bpm: a calm one for drafting and building, and a wave loop twice its length; they cross-fade at the same bar.
- Recorded CC0 hits with a synthesized layer under them; music ducks under big moments; kill streaks climb the pentatonic scale.

Monetisation (planned, not in the build)

- Rewarded ads first: a fourth card in a draft, an extra life when the gate falls on the last wave, double embers for a wave.
- Interstitials from level 4, between levels only. No paid currency until retention is known.

How it is built

- Unity 6 (URP), portrait, the new Input System; a uGUI interface with TextMeshPro.
- **Bonemarch.Core**: the rules in pure C# with no engine references, on a fixed 60 Hz step. The game, the EditMode tests and the balance runner all use it, so the bots play exactly the game players get.
- 71 EditMode tests (combat, draft, bots, tuning, save, levels and landmarks) and PlayMode tests that render every screen and capture the ad frames.
- Editor menus rebuild the scene and art from code, and ship the WebGL zip and the Android APK.

Roadmap

NEXT	WHY
Redesign the eight levels by hand	The drafts are a starting point; level 6's fork should split Necromancers from the troops they heal
Playtest levels 2 to 4	Real first-time players are probably weaker than the bot there
Decide the boons	They are near-neutral picks today
Analytics and a CPI test	See the ad and KPI plan

> **My role.** Concept, rules, the draft, the economy and the difficulty curve, the landmarks and the brief for every level, UX and art direction. Built in Unity (C#) with an AI coding assistant (Claude): I set the rules and the targets, reviewed every build and decided each change.